
COSMOS CONVERT

The Ultimate Document & File Conversion Suite

A premium SaaS and desktop platform comparable to Adobe Acrobat, Notion, and Canva.

Engineered with Next.js 15, FastAPI, Celery, Redis, SQLite/PostgreSQL, and Tauri.

Designed to process secure multi-format conversions, OCR extractions, and document styling.

CORE CAPABILITIES INCLUDED:

- Dynamic multi-format converter (PDF, DOCX, HTML, TXT, PNG, JPG, WEBP)
- Advanced PDF Toolbox (Merge, Watermark, Split, Protect, Decrypt, Rotate, Delete pages)
- AI OCR invoice parsing and business card extraction
- Interactive TipTap document rich text editor
- Stellar Image Editor sandbox with cosmic filters and crop bounds
- Multi-language system with flags (22 languages and RTL support)
- Galaxy theme backdrop visual settings (twinkling starfields, animated nebulae)

1. SYSTEM ARCHITECTURE & DEPLOYMENT

FastAPI Backend Microservice:

Serves REST API endpoints on port 8000. Utilizes SQLite in development for zero-dependency local setups, and automatically scales to PostgreSQL in production environments. CORS configuration enables communication from web browsers and Tauri desktop wrappers alike.

Celery Async Task Queue:

Handles compute-heavy document modifications in background threads to protect connection timeouts. Redis acts as the broker and task cache queue. If Redis is down, backend gracefully falls back to local synchronous execution.

Next.js 15 Web Portal:

Runs a premium client-side workspace on port 3000. Bootstrapped with Tailwind CSS, Framer Motion, and TipTap modules. Build configurations compile static assets for offline desktop apps.

Tauri Executable Wrapper:

Packages static web assets into highly optimized, native desktop apps (EXE, DMG, AppImage) with low memory footprints.

2. WORKSPACE OPERATIONAL GUIDE

Step 1: Demo Authenticate

Log in immediately using the '· Launch Quick Demo Workspace' button on the landing page, or authenticate with credentials user@cosmos.com / cosmos123. System automatically creates SQLite sessions.

Step 2: Upload Files

Drag and drop any file directly onto the dashboard container or browse select files. The system uploads and tracks storage quota sizes.

Step 3: Convert File Types

Choose an input document and target format from the selection dropdowns. Conversions process in real-time and list outputs in your history table.

Step 4: PDF Modifications

Perform mergers, watermarking overlays, passwords locking/decryption, rotation, and file size compressions. Converted files download directly.